

EE1101: Circuits and Network Analysis - Quiz 2

Name: _____

Roll No: _____

Instructions

1. The exam duration is 45 minutes, starting at 15:40. Ensure that you return the paper by 16:25. Please be aware that **late submissions** may incur **penalties**.
2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.
3. Adopting unfair practices will lead to an F grade and will be reported to the disciplinary committee.

I hereby declare that I have read all the instructions carefully and will adhere to them.

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1. (4 points) Consider the BJT circuit shown in Fig. 1(a). Using the equivalent circuit shown in Fig. 1(b), compute (assuming $r_b \rightarrow 0$ and $r_e \rightarrow 0$ and $\beta = 200$)

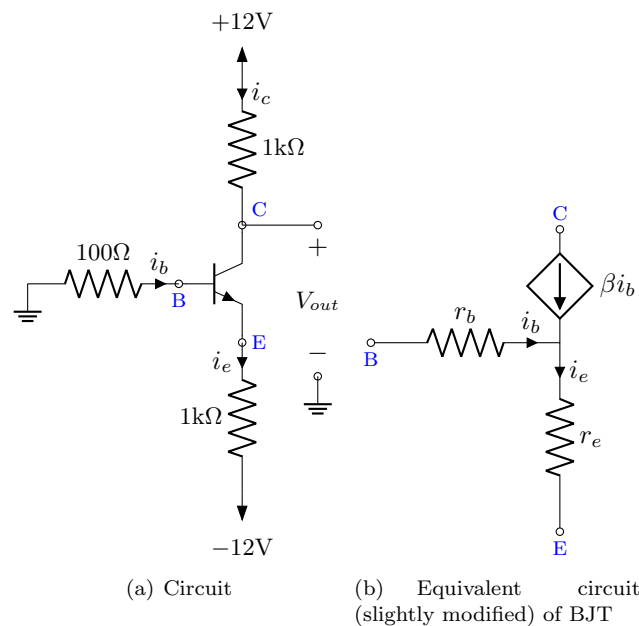


Fig. 1: Circuit with BJT (Questions 1)

- (2 points) The voltage V_{out} _____.
 - (2 points) The collector current i_c _____.
2. (4 points) For the signal shown in Fig. 2, compute
 - (1 point) the period (T) _____.

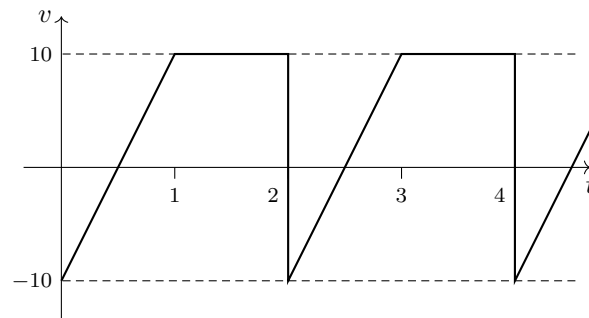


Fig. 2: Signal for Analysis (Question 2)

- (b) (1 point) the average value (full cycle) _____.
- (c) (2 points) the root mean square value _____.
3. (2 points) For the circuit shown in Fig. 3, compute the current through 1Ω resistor.

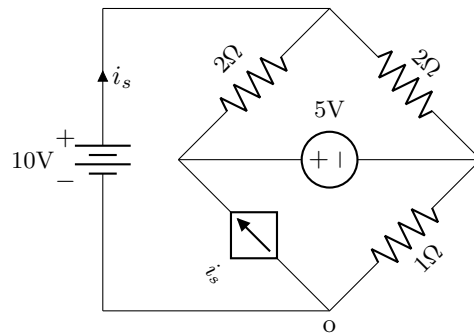


Fig. 3: Circuit for question 3

The current through 1Ω resistor is _____.

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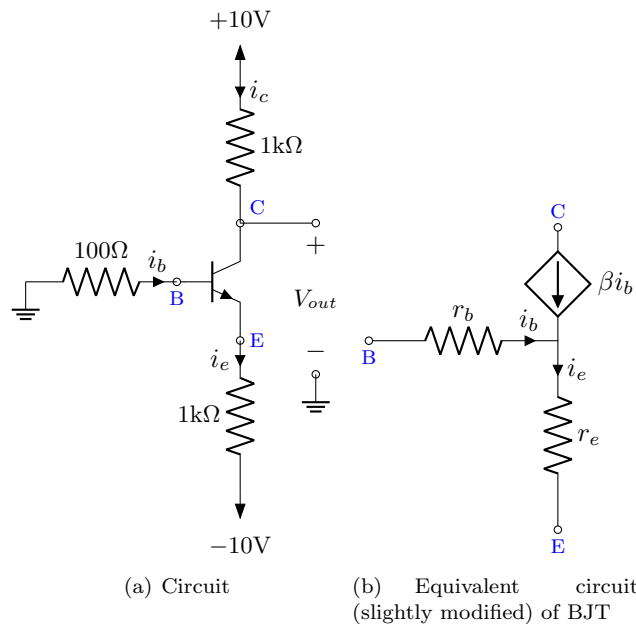


Fig. 1: Circuit with BJT (Questions 1)

- (2 points) The voltage V_{out} _____.
 - (2 points) The emitter current i_e _____.
2. (4 points) For the signal shown in Fig. 2, compute
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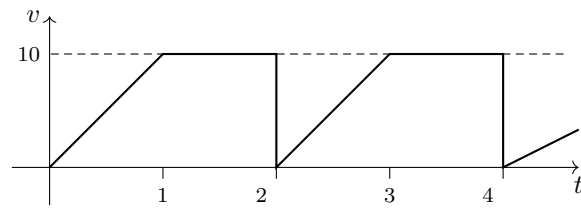


Fig. 2: Signal for Analysis (Question 2)

(b) (1 point) the average value (full cycle) _____.

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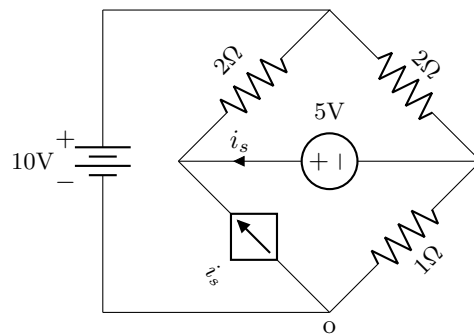


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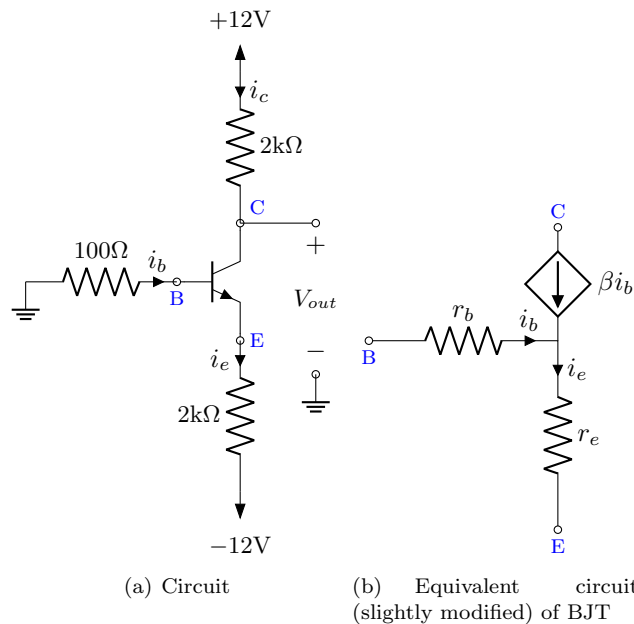


Fig. 1: Circuit with BJT (Questions 1)

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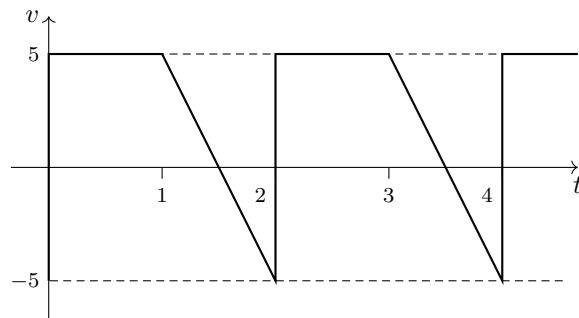


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- (b) (1 point) the average value (full cycle) _____.
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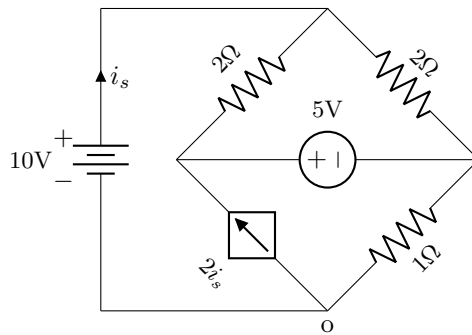


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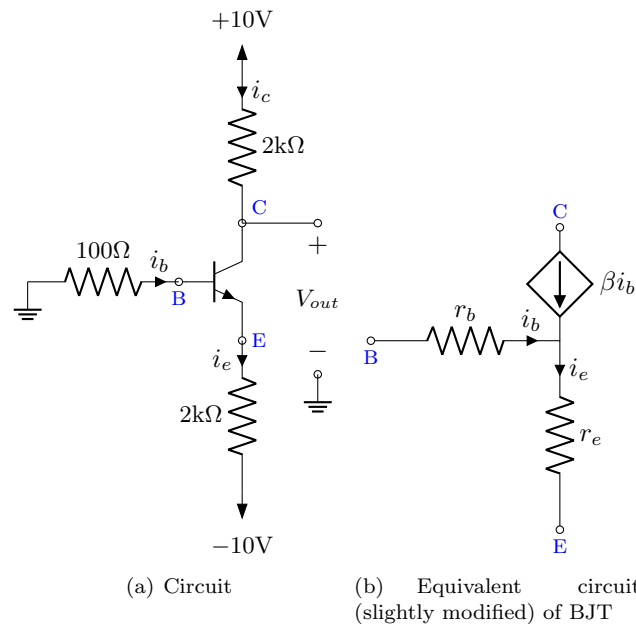


Fig. 1: Circuit with BJT (Questions 1)

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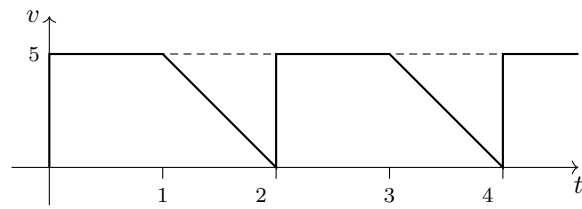


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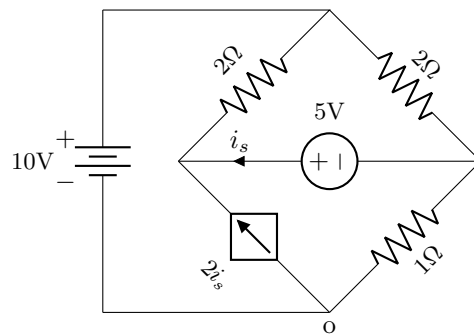


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